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Paleomagnetism of the Structurally Deformed Latir Volcanic Field, Northern New Mexico: Relations to Formation of the Questa Caldera and Development of the Rio Grande Rift

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The Oligocene-Miocene Questa caldera and related rocks of the Latir volcanic field are exposed along the eastern flank of the Rio Grande rift in northern New Mexico. These rocks bracket in age the beginning of extensional tectonics in the region approximately 25–26 m.y. ago. The early phase of extension is characterized by currently low-angle normal faults that caused steep tilting of volcanogenic units, particularly within the Questa caldera. Paleomagnetic measurements on intrusive and extrusive rocks from the Latir field show that tilting began abruptly after the eruption of the Amalia Tuff and subsequent formation of the Questa caldera 26 m.y. ago. Specifically, most of the tilting occurred after the intracaldera welded tuff and uppermost portion of the subcaldera magma chamber had cooled and become magnetized, but before the subsequent resurgent plutons had been emplaced and cooled sufficiently to also become magnetized about 1 m.y. later. Differential uplift of the region exposes similarly aged subvolcanic plutonic rocks to the south that also have untilted magnetizations. Tilting had ceased altogether by the time the late-stage plutons were magnetized, at least within the caldera structure and to the south. Younger rift-related sedimentary rocks and interbedded basalt flows (15 m.y. old) show that tilting continued to the north of the caldera past this time. However, paleomagnetic evidence indicates that a nearby 7- to 11-m.y.-old rhyolite dome is probably untilted, suggesting that significant tilting in the region had ended by this time. The latest phase of extension is characterized by high-angle deeply penetrating normal faults, which are responsible for the high structural relief presently observed in the region.

INTRODUCTION

Uplift and erosion of the Sangre de Cristo Mountains of northern New Mexico along the eastern flank of the Rio Grande graben have exposed in cross section the Oligocene-Miocene Questa caldera and related rocks of the Latir volcanic field (Figure 1). These rocks comprise intermediate- to rhyolitic-composition volcanics, epizonal intrusives, and associated mesozonal plutonic rocks that bracket the inception of, and have been disrupted by, intense rift-related faulting [*Lipman*, 1981]. The eruption of the Latir volcanic rocks and emplacement of the subvolcanic intrusive rocks appear to have occurred within a period of a few million years, including the eruption of a single regional ash flow sheet and the concomitant formation of the Questa caldera [*Lipman*, 1983]. The caldera structure is complex and differs significantly from typical caldera models [e.g., *Smith and Bailey*, 1968] due primarily to its disruption by extensional faulting related to the Rio Grande rift [*Lipman*, 1983].

This study concerns the application of paleomagnetic techniques to the analysis of structural deformation within the Latir volcanic field in order to better understand the relationship between regional mid-Tertiary igneous and tectonic events. A comparison of paleomagnetic directions from the extrusive rocks of the Latir field with the expected mid-Tertiary geomagnetic field direction for northern New Mexico provides (1) an independent check of their internal "bedding" features (presumed to indicate the paleohorizontal) that apparently indicate intense structural deformation, and (2) information on attitudes for exposures that contain no consistent indication of the paleohorizontal. In the intrusive units, paleomagnetic results give the only available postcooling attitude for these rocks. The paleomagnetic data presented in this paper include and build upon those presented in an earlier report [*Hagstrum et al.*, 1982]. These results have been reanalyzed to conform with the component selection techniques applied to the more recently obtained data. The preliminary interpretation, although basically correct, has been expanded and revised to incorporate the new data. The new results, which represent approximately two-thirds of the total sample collection, come from previously unsampled units (mostly in the subvolcanic intrusives), from geographically dispersed localities in previously sampled rock types, and from those areas initially giving conflicting or confusing results.

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TECTONIC AND GEOLOGIC SETTING

The Rio Grande rift represents a northeastward extension of the southern Basin and Range province, separating the stable cratonic Colorado Plateau and southern High Plains regions. The Rio Grande rift initially developed along a major zone of crustal weakness associated with late Paleozoic and Late Cretaceous/early Tertiary (Laramide) tectonism [*Chapin*, 1979; *Cordell*, 1978]. Widespread intermediate- to silicic-composition volcanism, related to near-surface batholith emplacement, preceded rifting and began around 40–35 m.y. ago in the southern Rocky Mountains of Colorado and northern New Mexico [*Lipman et al.*, 1972] and around 43–40 m.y. ago in the Mogollon-Datil field of southwestern New Mexico

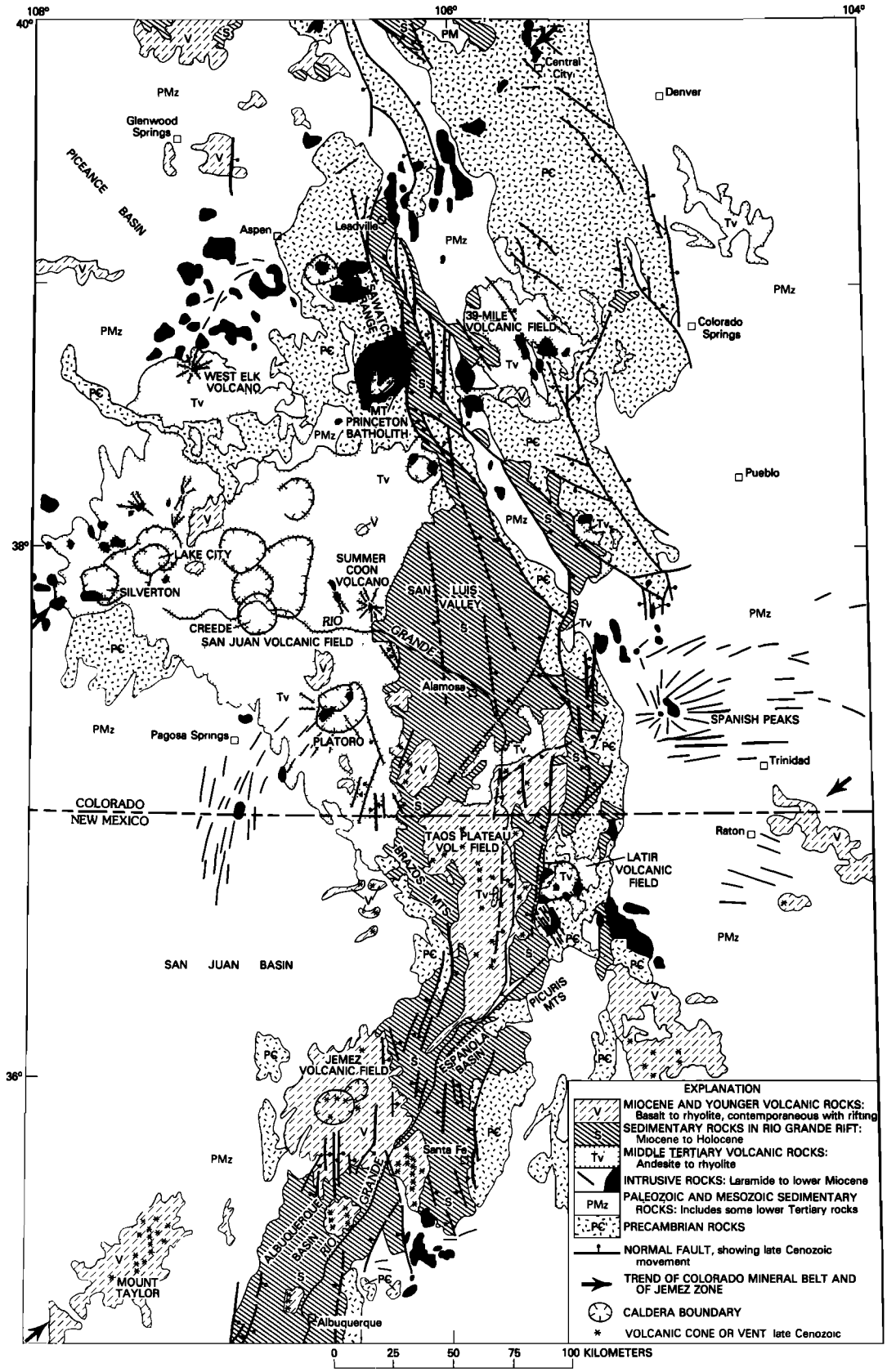


Fig. 1. Index map of southern Rocky Mountains showing regional setting of the Latir volcanic field in northern New Mexico. Modified from *Tweto* [1978] and *Woodward et al.* [1978].

[*Elston et al.*, 1976; *Elston and Bornhorst*, 1979]. Extensional faulting, bolson sedimentation, and major volumes of basaltic andesite volcanism characterized the early extension, which started about 29 m.y. ago in the Socorro–Las Cruces area to the south [*Chapin and Seager*, 1975; *Chapin*, 1979] and around 26 m.y. ago in the San Luis Valley to the north [*Lipman and Mehnert*, 1975]. Contemporaneous with the beginning of extension in these regions was the eruption of high-

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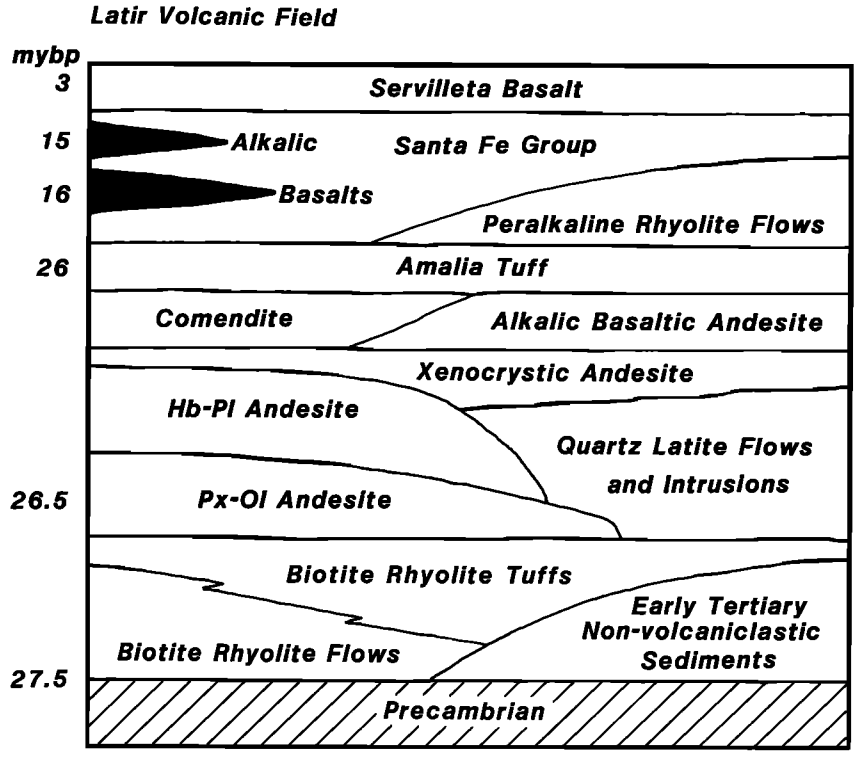


Fig. 2. Schematic stratigraphic section of the Latir volcanic rocks. Apparent thickness is unrelated to the true thickness of the rock unit.

silica alkalic rhyolitic rocks from large volcanic centers [*Elston et al.*, 1976; *Lipman et al.*, 1978].

Early-rift volcanic rocks exposed in the Lemitar Mountains (an intrarift fault block) west of Socorro, New Mexico, were intensely faulted and tilted (up to 60°) during the early extension. Movement along closely spaced, down-to-the east, currently low-angle, normal faults caused most of the tilting between about 29 and 20 m.y. ago, whereas subsequent deformation along high-angle faults produced less tilting of the strata [*Chamberlain*, 1978]. A regional lull in volcanism, faulting, and uplift followed during the mid-Miocene (20–13 Ma) in the southern Rio Grande rift [*Chapin and Seager*, 1975]. Volcanism returned slowly after this period of relative inactivity until a rapid acceleration in basaltic volcanism occurred about 5 m.y. ago, which accompanied uplift and block faulting in the southern Rocky Mountains [*Chapin*, 1979]. Uplift and extension have persisted to the present day, as shown by recent fault scarps, high heat flow, and geophysical evidence of modern magma bodies and anomalous crust and upper mantle beneath the rift [*Chapin*, 1979; *Cordell*, 1978].

The Latir volcanic field and associated intrusive rocks form the southeasternmost portion of a large composite Tertiary volcanic field that developed across older eroded Laramide uplifts in the southern Rocky Mountain region [*Steven*, 1975, Figure 1]. The general sequence of volcanic rocks in the Latir field (Figure 2) is similar to that in other portions of the larger field: rhyolites and then predominantly intermediate-composition lavas and breccias are overlain by silicic ash flow tuffs. The ash flow tuffs are associated with caldera collapse and were followed by emplacement of granitic rocks in the vent regions. Early mapping of the Latir volcanic field established the basic distribution of the major rock types [*McKinlay*, 1956, 1957; *Clark and Read*, 1972], but more recent mapping by *Lipman* [1983] and *Reed et al.* [1983] has refined the stratigraphy, identified the margins of the Questa caldera, and recognized a high degree of extensional faulting within the volcanic units.

The initial phase of volcanic activity in the Latir field (28–26 Ma, Table 1) comprised the eruption of mostly andesitic lavas and breccias from large central volcanoes. Surrounding these volcanoes, and interlayered with the flows, were alluvial deposits of volcaniclastic debris. At or near the Precambrian-Tertiary unconformity, however, low-silica rhyolitic rocks occur as rhyolite flows and as domes with flanking ash flow tuffs (Figure 2). Interbedded with, but generally overlying, the andesitic lavas are the thick flows and domes of the Latir Peak Quartz Latite. Small stocks northwest of the Questa caldera and large laccolithic bodies of quartz latite to the south were also intruded at about this time. The subsequent caldera-forming eruption of the Amalia Tuff (26 Ma) was accompanied by similar-composition rhyolitic lava flows and intrusions. The tuff was emplaced over an irregular topography and ponded within the collapse structure. Tilting of the volcanic units also occurred at about this time [*Lipman*, 1983]. This early phase of rifting in the Questa region was characterized by normal faults, which varied widely in trend, and by steep tilting of strata (locally near vertical within the caldera structure), although the dominant trend of faults across this region suggests a NE-SW directed extension [*Lipman*, 1981].

Postcollapse resurgent plutons elevated and intruded the caldera floor 25–26 m.y. ago. Although the resurgent plutons are primarily silicic granites, the slightly older pluton of Virgin Canyon (Table 1) shows weakly peralkaline affinities that suggest it may be an intrusive equivalent of the Amalia Tuff [*Lipman*, 1983]. Three late-stage plutons, presumed to connect at depth, intruded the caldera wall, floor, and fill along the southern margin about 22–23 m.y. ago. A similar high-silica granite was also emplaced at Lucero Peak, south of the caldera, around 21–22 m.y. ago. Associated with these granitic

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TABLE 1. Ages for the Tertiary Igneous Rocks of the Latir Volcanic Field Interpreted From Available Potassium-Argon and Fission Track Determinations

Unit	Age, m.y.	Reference
Postcaldera Volcanics		
Servilleta Basalt	3–4	1
Rhyolite dome at Gonzales Ranch	7–11	2
Basalt flows (Santa Fe Group)	15–16	2
Intracaldera Plutons		
Cabresto Lake	25–26	2
Rito del Medio	25–26	2
Virgin Canyon	26	2
Southern Caldera Margin Plutons		
Sulfur Gulch	22–23	3
Bear Canyon	22–23	3
Mineralization		
Sulfur Gulch	23	3, 4, 5
Bear Canyon	23	5
Southern Plutons		
Lucero Peak	21–22	2
Rio Hondo	25–26	2
Caldera-Related Rocks		
Amalia Tuff and cogenetic lavas	26–27	2
Precaldera Intrusive and Extrusive Rocks		
Comendite at Ortiz Peak	26	2
Latir Peak		
Quartz Latite	26–27	2
Andesitic and rhyodacitic flows	27–28	2
Tuff of Tetilla Peak	27–28	2

All K-Ar ages corrected to current constants. 1, *Lipman and Mehnert* [1979]; 2, *Lipman et al.* [1986]; 3, *Laughlin et al.* [1969]; 4, *Shassberger* [1972]; and 5, *Shibata and Ishihara* [1968].

intrusives is molybdenite mineralization deposited from hydrothermal fluids derived from the silicic magmas [*Carpenter*, 1968; *Leonardson et al.*, 1983]. Attendant alteration and propylitization of the host volcanic rocks are particularly intense near the southern margin of the Questa cladera but are reduced in intensity toward the perimeter of the volcanic field. Paleomagnetic evidence shows that these late-stage plutons became magnetized after the tectonic deformation of the caldera area was complete [*Hagstrum et al.*, 1982].

Differential uplift of the Taos block of the Sangre de Cristo Mountains has exposed deeper structural levels to the south of the Questa caldera, uncovering the large Rio Hondo pluton (Figure 3). The pluton consists predominantly of quartz monzonite, which grades upward into high-silica granite near its roof. Contacts at the roof generally have gentle dips, and the steep linear contact along the pluton's northeast side is localized along a major NW trending fault [*Lipman*, 1983]. Hundreds of NW trending porphyritic to aphanitic dikes cut the body and range in composition from basalt to rhyolite. Mostly of rhyolitic composition, the dikes are probably cogenetic with the 25- to 26-m.y.-old Rio Hondo pluton [*Lipman*, 1983] and demonstrate the early Miocene orientation of extension in the region [*Lipman*, 1981].

The Santa Fe Group sediments overlie the volcanic rocks of the Latir field, locally include interbedded basalt flows dated at 15–16 Ma (Table 1), and accumulated in fault-bounded basins that developed as extension continued. The Santa Fe Group rocks dip to the northeast, indicating that tilting persisted after 15 Ma to the north of the caldera region. Capping these rocks is the untilted 3- to 4-m.y.-old Servilleta Basalt, which has been uplifted approximately 500 m relative to counterparts in the San Luis Valley [*Lipman and Mehnert*, 1979]. The most recent north trending, high-angle, normal faults have formed a deep narrow graben along the eastern margin of the present San Luis Valley, downdropping and burying the western portion of the Questa caldera beneath the valley fill (Figure 3).

FIELD AND LABORATORY TECHNIQUES

A total of 1038 independently oriented samples were collected from 131 sites either as drill cores taken by means of a portable gasoline-powered rock drill or as hand samples. Orientations of both drill cores and hand samples were accomplished with a magnetic and/or solar compass and clinometer. In the laboratory, cores similar to those collected in the field (2.5 cm in diameter) were drilled from the hand samples, and specimens approximately 2.2 cm in length were cut from all cores. Accuracy of orientation for the samples is estimated to be $\leq 2°$ for field-drilled cores and $\leq 5°$ for hand samples.

Samples were collected throughout the Tertiary units of the Latir field from geographically dispersed sites (Figure 3) to avoid or delimit suspect zones of remagnetized rocks, particularly in close proximity to the youngest granitic intrusions. Because of limited exposures and structural complexity, the stratigraphic order in some intermediate-composition volcanics is uncertain. In general, exposures on ridge crests and cliff tops were avoided to guard against spurious lightning-induced magnetizations [*Graham*, 1961; *Cox*, 1962]. However, in a few cases, samples were obtained near ridge crests when other useful exposures were unavailable. Structural corrections for tilt were derived in the extrusive units from flow features and contacts, eutaxitic structures, and interbedded sediments. Several attitude measurements were usually averaged across the rock exposure at a site, but in many cases, no direct structural correction was available. The structural corrections reported here were made by simple rotation of the measured paleohorizontal plane to the present horizontal about the strike axis. Dispersion in the declination of the magnetic directions may therefore be introduced by unrecognized local plunge or rotations of structure [*MacDonald*, 1980].

The remanent magnetization of individual specimens was measured with a cryogenic magnetometer. All samples were subjected to either progressive alternating field (AF) or thermal demagnetization, averaging about 8 steps per specimen. AF demagnetization to peak fields of 70–100 mT was the primary technique used in this study and was carried out using a tumbling AF demagnetizer. Thermal demagnetizations were performed in air to temperatures usually in excess of 600°C, employing a shielded, noninductively wound furnace. Components of magnetization were selected from the demagnetization data using a least squares method based on principal component analysis modified from *Kirschvink* [1980]. Either lines (single components) or planes (two overlapping components) were fitted to the magnetization vector end points in three-dimensional space. Demagnetization planes (remagnetization circles) were used for those sites where the primary and secondary components of magnetization overlapped and could not be directly resolved. The statistical

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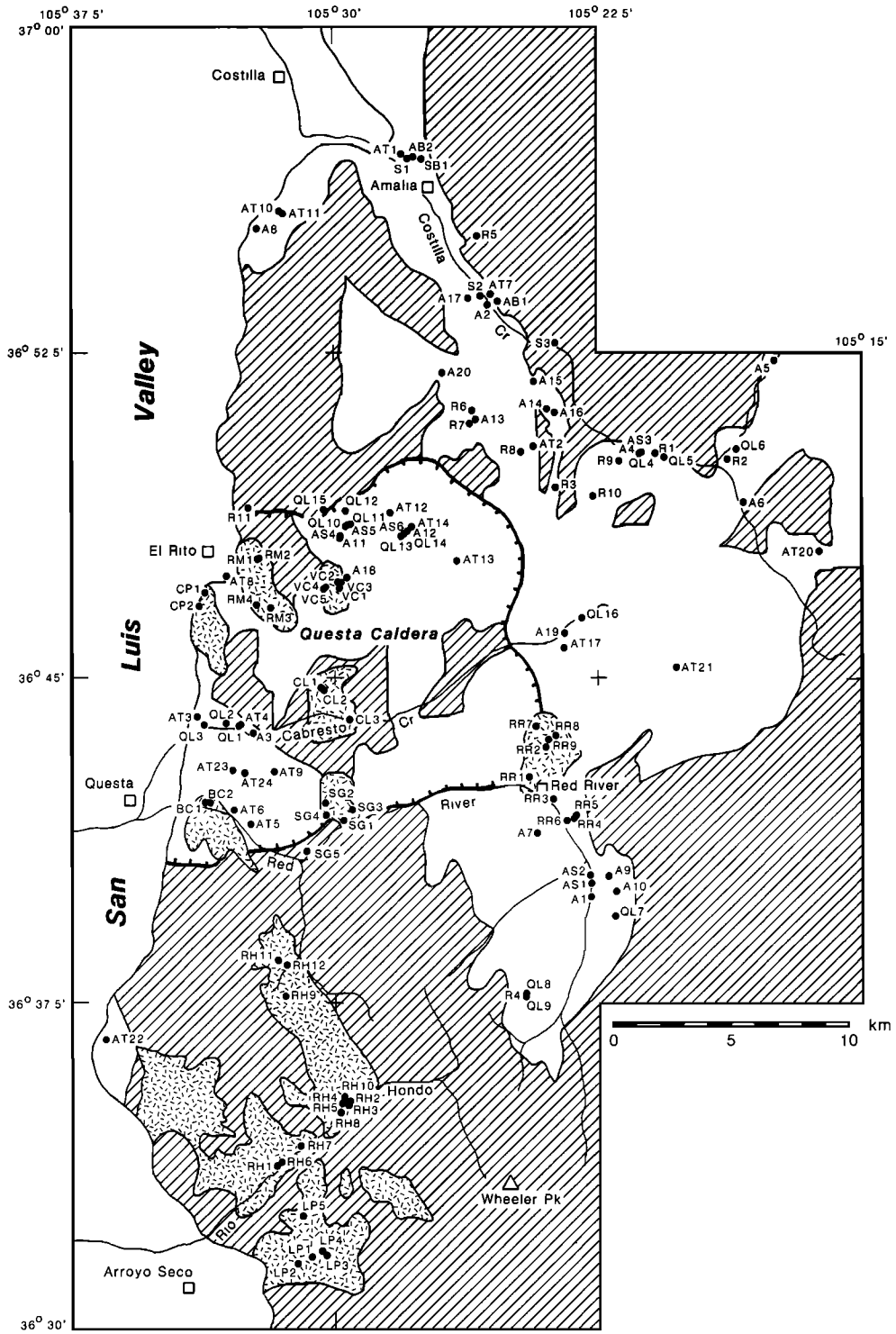


Fig. 3. Simplified geologic map of the latir volcanic field and Questa caldera showing locations of the paleomagnetic sampling sites. Shaded areas represent pre-Tertiary, mostly Precambrian, rocks, and unshaded and stippled areas east of the San Luis Valley represent the Tertiary volcanic and intrusive rocks of the Latir field, respectively. Site prefixes denote rock units as follows: A, andesites; AS, interbedded sedimentary rocks; QL, quartz latites; AT, Amalia Tuff; R, rhyolites; S, sedimentary rocks of the Santa Fe Group; AB, intercalated basalts; SB, Servilleta Basalt; RH, Rio Hondo pluton; LP, Lucero Peak pluton; SG, Sulfur Gulch pluton; BC, Bear Canyon pluton; RR, Red River plutonic complex; CL, Cabresto Lake pluton; RM, Rito del Medio pluton; VC, Virgin Canyon pluton; and CP, Cañada Pinabete pluton.

methods of *Fisher* [1953] are used in analyzing the within-site line directions and unit means, and Bingham statistics [*Onstott*, 1980] were used to calculate the mean direction for those sites exhibiting intersecting demagnetization planes. Occasionally one or two sample-mean directions that gave directions well away from the site-mean direction were discarded from a site-mean calculation. These samples either showed unstable demagnetization behavior or had directions attribu-

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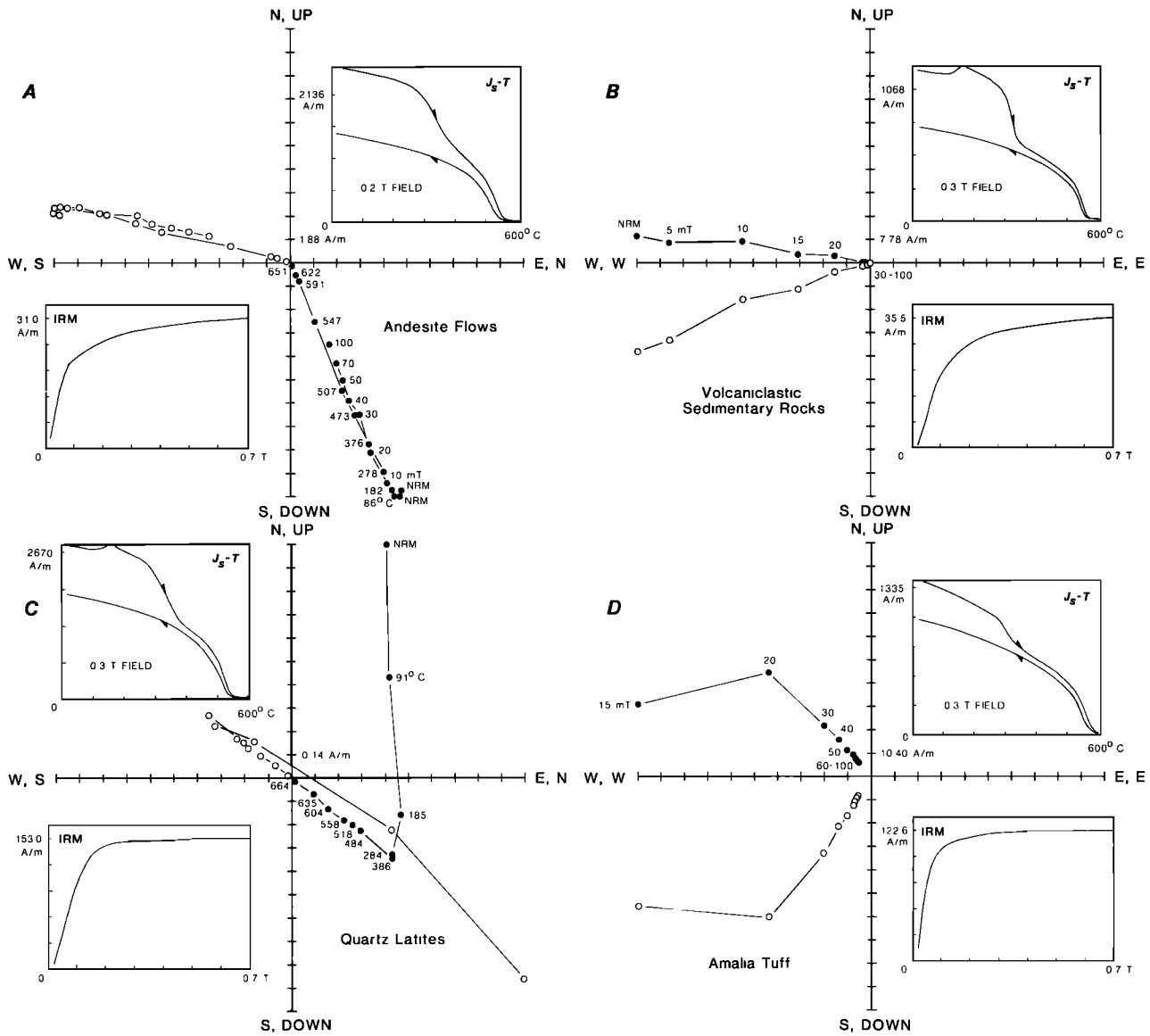


Fig. 4. Orthogonal projections (uncorrected for tilt of bedding) of demagnetization sequences for samples from the volcanic units of the Latir field. (*a*) AF and thermal results from a single sample in the andesite flows from site A15. Insets exhibit J_s-T and IRM results on small cores from the same site. (*b*) AF results from site AS5 in the volcaniclastic sedimentary rocks interbedded with the andesite flows. (*c*) Thermal results on a quartz latite specimen from site QL8. (*d*) AF result from site AT14 in the Amalia Tuff. Solid circles, projections of vector end points onto horizontal plane; open circles, onto vertical plane.

table to lightning or overprint-related effects, observed in other samples from the same site, that could not be resolved with demagnetization or remagnetization circle analysis.

In order to define better the magnetic mineralogy, Curie temperatures were measured on 0.6-cm-diameter cores from representative specimens with a recording magnetic balance that determined strong field magnetization as a function of temperature (J_s-T). These small cores were heated to 620°C and then cooled in steady fields of 0.2–0.5 T at rates of 10°C/min in an Ar atmosphere. The strong field method of determining Curie temperatures is effective for titanomagnetite compositions, but the method cannot detect hematite even in the presence of small quantities of titanomagnetite. Thus the progressive acquisition of isothermal remanent magnetization (IRM) in fields up to 0.7 T was measured on selected small cores as a means for detecting hematite mixed with appreciable amounts of titanomagnetite. Also, polished thin sections from representative sites were examined to identify the magnetic constituents.

RESULTS

Andesitic Flows and Related Sedimentary Rocks

The 27- to 28-m.y.-old andesitic lava flows were sampled at 20 sites, and the interbedded volcaniclastic sedimentary rocks at six sites throughout the Latir field (Figure 3). At most sites in the flows, structural attitudes for the flows were taken from interlayered sedimentary strata where present, or from the nearest stratigraphically related field determination to the sampling site. Each site is considered to represent a single lava flow, i.e., a spot reading of the ancient geomagnetic field.

Natural remanent magnetization intensities (J_{NRM}) for the

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andesites, prior to laboratory treatment, ranged from approximately 10 to 10^{-1} A m^{-1}. Upon demagnetization, these rocks consistently displayed univectorial behavior (linear decay of vector end points toward the origin) after the removal, usually by 30 mT or 300°C, of any low-coercivity (H_c) or low-blocking temperature (T_b) spurious magnetizations. For most sites, a significant portion of the magnetization (20–40% of J_{NRM}) remained after AF demagnetization to 100 mT, and at two sites (A2 and A18), 70–80% of the J_{NRM} value remained after the 100-mT step. Thermal demagnetizations to 650° or 680°C were performed on these high-H_c samples, yielding complete demagnetization of univectorial components identical to those observed with AF treatment (Figure 4*a*). Samples from sites A3, A8, A16, and A17 were demagnetized to 5–10% of their J_{NRM} values with alternating fields. These results suggest that the primary carrier of the remanent magnetization in the andesites is titanomagnetite, which is further supported by IRM and J_s-T determinations (Curie temperatures (T_c) of 554°–590°C), and polished thin-section observations. In the samples where AF analysis proved less effective in completely demagnetizing the NRM, subordinate amounts of secondary hematite also contribute to the magnetic remanence. For many sites, the J_s-T measurements indicate the presence of maghemite by a pronounced decrease in magnetization between 300° and 400°C upon heating and by an overall reduction in magnetization upon cooling (Figure 4, insets). In a few instances, maghemite is confirmed by reflected-light observations, but generally the magnetic grains are too small for clear identification. In addition, a few specimens yield J_s-T heating curves having sharp increases in magnetization between 150° and 165°C (Figures 4*b* and 4*c*). This "dorsal fin" on the heating curve indicates the presence of minor pyrrhotite for compositions ranging from Fe_9S_{10} to FeS [*Schwartz*, 1975]. Because such pyrrhotite compositions are antiferromagnetic at room temperature, the mineral does not contribute to the NRM.

AF demagnetization to 100 mT or thermal demagnetization to 590° or 650°C of samples from the interbedded volcaniclastic sedimentary rocks was successful in reducing the J_{NRM} values to low levels (< 10%), except at site AS6, in which 50–60% of the J_{NRM} remained after AF treatment to 100 mT and approximately 30% remained after heating to 650°C. With the exception of this one site, AF demagnetization proved to be the most effective method in resolving the stable magnetizations in these sedimentary rocks (Figure 4*b*). Univectorial behavior was exhibited at most sites, but at AS3, AS4, and AS5, some samples showed secondary components that persisted to high levels of demagnetization. The IRM results, in support of the demagnetization evidence, show that the stable remanence of the volcaniclastic sedimentary rocks, like the andesites, resides in titanomagnetite accompanied by some hematite.

Within-site dispersion of the sample directions for the andesites is small (Table 2), and for sites in the volcaniclastic sedimentary rocks the within-site dispersion is somewhat greater (Table 2). For all but one site in the flows and sedimentary rocks, components of magnetization are estimated by least squares fits to the demagnetization vector end points. Samples from site A16 in the xenocrystic andesite have converging great-circle arcs on equal-area plots, indicating the simultaneous removal of a set of random magnetizations that overprint a consistent higher H_c and T_b magnetization. Thirty to forty percent of the total sample collection was discarded from sites A8, AS4, and AS5, as these samples gave directions of magnetization diverging more than 40° from the site mean calculated from the remaining samples. These magnetizations are considered to be strongly influenced by lightning-induced components because they generally show significantly higher intensities, are widely scattered in direction, and come from topographically high and exposed locations.

The site means for the andesite flows and associated sedimentary rocks are generally to the southeast with shallow to moderate inclinations for reversed polarity directions (Figure 5), and to the west-northwest with similar inclinations for normal polarity directions, significantly different from the predicted mid-Tertiary field direction (Table 3). These directions are consistent with the structural trends observed in the Latir volcanic field [*Lipman*, 1983], where the extrusive units are variably tilted to the northeast. The stability of the remanent magnetization is best demonstrated by a "fold" test on the interbedded sedimentary layers (AS sites), which give the most reliable measure of the paleohorizontal. Because these sites are not from a single stratigraphic horizon and require individual corrections, a rigorous test [*McFadden and Jones*, 1981] cannot be applied. However, a comparison of the Fisher precision parameter **k** of the averaged site-mean directions before and after corrections for each site shows an increase from 5 to 16, significant at the 95% confidence level [*Watson and Irving*, 1957]. The magnetization of the sedimentary rocks therefore probably predates the structural deformation. A similar comparison for the andesite site-mean directions, using the best available corrections, also shows an increase in **k** (4 to 7) that is just significant at the 95% level. Again, the magnetizations most likely predate the structural disturbance of these rocks, but the angular dispersion (**s** = 30.6°) after correction is large (see discussion), suggesting that these structural corrections do not completely represent the deformation of the andesites.

Some of the andesites (sites A2, A3, and A7), with uncorrected directions close to the normal polarity predicted field direction were considered by *Hagstrum et al.* [1982] to have been remagnetized by the granitic intrusions postdating the structural deformation of the area. In the present data set, it is difficult to assess the possibility of remagnetization on the basis of magnetic directions alone, especially for sites with small structural corrections, where the effect of secular variation could be greater than the structural correction. For unremagnetized rocks, this could result in the corrected direction at times being further from the predicted direction than the uncorrected direction. In this study, a site is considered remagnetized when its mean direction moves from a position near the predicted field direction to one well away from it ($\geq 40°$, see discussion) upon application of the assigned structural correction and the site is in close proximity to subsequent igneous activity and alteration. Using these criteria, the most likely candidate for remagnetization among the andesites is site A3.

Latir Peak Quartz Latite and Similar-Composition Intrusive Rocks

The 26- to 27-m.y.-old quartz latitic rocks were emplaced as massive porphyritic gray- to red-brown lava flows, domes, or hypabyssal intrusions. These rocks were sampled at 16 sites in the Latir field (Figure 3). Structural corrections were estimated for the quartz latites from the closest stratified units (typically

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TABLE 2. Site-Mean Directions and Poles for the Extrusive Units

	Uncorrected			Corrected						Pole	
Site	*I*	*D*	Stk/Dp	*I*	*D*	*N*	*R*	*k*	α_{95}	Latitude	Longitude
				Andesite Flows and Interbedded Sedimentary Rocks							
A1	−15.6	133.6	14/35	−44.4	145.8	8	7.9627	188	4.1	−59.1	334.5
A2	56.8	27.9	357/38	29.1	54.5	8	7.9686	223	3.7	37.5	353.4
A3	39.0	41.2	172/80	43.5	306.5	6	5.8673	38	11.1	43.4	166.2
A4	−12.8	140.7	2/40	−35.7	156.4	7	6.9610	154	4.9	−63.3	311.5
A5	−32.8	113.3	359/40*	−65.2	144.3	8	7.9674	215	3.8	−61.8	17.6
A6	−27.1	187.5	8/25*	−24.6	199.8	9	8.9561	182	3.8	−60.2	213.0
A7	42.7	333.8	12/35*	54.7	13.6	9	8.6518	23	11.0	78.9	348.2
A8	−74.1	187.2	319/15*	−61.2	206.7	7	6.9641	167	4.7	−68.8	141.7
A9	−35.0	146.0	331/40*	−29.0	172.0	9	8.9248	106	5.0	−67.7	275.3
A10	−2.0	120.6	331/40*	−20.6	128.1	8	7.8441	45	8.4	−36.6	329.2
A11	−10.0	280.3	345/80	57.9	307.3	6	5.9038	52	9.4	48.8	183.5
A12	44.6	304.0	335/50	47.1	1.3	6	5.8716	39	10.9	81.4	66.9
A13	−6.4	106.6	331/50*	−37.2	124.0	7	6.9959	1478	1.6	−39.1	342.4
A14	−33.8	134.3	330/40*	−34.8	163.0	7	6.9777	269	3.7	−66.9	299.3
A15	−12.2	156.9	15/15*	−21.1	160.6	8	7.9810	369	2.9	−58.7	293.6
A16	−49.9	55.1	290/20*	−64.0	77.7	7	6.9442	108	6.8	−18.1	28.8
A17	57.7	356.3	353/35*	42.5	36.7	7	6.9649	171	4.6	56.4	355.5
A18	−48.7	92.6	339/30*	−72.3	129.8	6	5.9775	222	4.5	−51.3	33.2
A19	46.2	310.4	330/40*	44.6	353.7	7	6.9932	880	2.0	78.2	103.2
A20	22.6	236.2	330/40*	62.4	232.4	10	9.8132	48	7.0	3.6	219.6
Avg	39.7	313.2		51.8	341.9	20	17.2390	7	13.4	73.7	164.9
AS1	51.8	352.0	2/20	50.8	17.4	6	5.9102	56	9.1	74.7	359.2
AS2	50.6	1.6	32/18	56.6	24.7	5	4.8385	25	15.7	70.3	335.6
AS3	57.1	314.8	2/40	64.1	34.4	6	5.9180	61	8.6	62.9	314.5
AS4	25.6	270.6	345/80	68.4	33.6	8	7.6079	18	13.5	62.1	301.6
AS5	−3.9	272.9	320/90	47.0	314.2	6	5.6407	14	18.6	50.8	165.4
AS6	31.8	299.1	335/50	46.1	341.8	6	5.8671	38	11.1	72.1	138.4
Avg	41.3	304.2		59.3	3.2	6	5.6907	16	17.2	83.9	300.3
				Quartz Latite Flows and Intrusives							
QL1-2	46.8	1.9	358/107*	−14.9	43.0	5	4.9766	171	5.9	81.1	63.5
QL2-2	48.4	357.0	358/107*	−12.0	45.3	5	4.9471	76	8.9	82.2	94.3
QL3-2	30.5	359.6	358/107*	−9.9	27.1	8	7.6452	20	12.8	69.7	75.4
QL4	29.0	296.8	2/40*	61.9	323.1	9	8.9199	100	8.1	61.3	188.8
QL5	23.6	286.7	2/40*	61.3	300.9	6	5.9605	127	7.2	45.0	190.9
QL5-2	57.2	5.7	2/40*	38.4	48.4	8	7.8540	48	8.1	85.4	331.0
QL6	26.1	290.8	2/40*	62.1	310.3	8	7.9831	414	4.0	52.0	190.4
QL7	−42.5	48.6	331/40*	−78.6	7.8	8	7.8112	37	13.3	−14.9	71.6
QL7-2	42.7	1.6	331/40*	16.2	19.8	8	7.6752	22	12.2	78.0	67.7
QL8	−25.3	135.0	335/50*	−30.7	163.6	9	8.8628	58	10.6	−64.5	293.4
QL8-2	58.2	358.5	335/50*	22.7	33.4	7	6.7849	28	11.6	88.3	211.6
QL9	50.1	287.2	335/50*	59.0	8.2	5	4.8980	39	12.4	83.2	323.5
QL10	7.9	277.5	345/80*	67.6	341.6	8	7.7467	28	10.7	71.0	216.5
QL11	−17.4	273.8	320/90*	43.5	295.7	7	6.8831	51	8.5	34.8	172.1
QL12	3.3	66.4	320/90*	73.3	308.5	8	7.8913	64	6.6	50.4	215.3
QL13	10.4	299.1	335/50*	33.9	318.7	5	4.9005	40	10.9	49.5	149.2
QL14	41.5	284.9	335/50*	60.0	351.0	6	5.8914	46	10.0	81.9	197.5
QL15	44.4	262.5	320/90*	37.1	21.2	8	7.7073	24	11.6	65.5	19.7
QL16	49.6	299.5	335/50*	51.0	7.9	7	6.9611	154	4.9	81.7	19.7
Avg	32.0	289.0		60.1	335.2	13	11.9542	11	12.8	69.0	192.7
Avg 2	47.3	0.6				6	5.9180	61	8.7	82.3	70.1
				Amalia Tuff							
AT1	46.4	334.1	5/31	53.4	11.7	8	7.9692	227	3.7	80.0	358.6
AT2	28.2	293.0	342/70	51.8	2.5	12	11.7619	46	6.4	85.2	48.7
AT3-2	67.0	19.1	358/107	−23.8	64.5	11	10.8961	96	4.7	71.1	295.2
AT4-2	39.3	352.2	347/115	−19.4	22.2	5	4.8372	25	15.8	74.0	101.5
AT5-2	19.1	26.9	302/65	−45.6	25.1	7	6.8289	35	10.3	53.8	25.5
AT6-2	14.1	4.9	2/90	−2.8	16.1	5	4.9543	88	8.2	60.1	64.6
AT7	33.0	319.6	357/40	48.2	353.5	8	7.9824	398	2.8	80.6	111.4
AT8-2	50.3	49.6	20/90	−18.4	74.2	6	5.8825	43	10.4	48.9	337.4
AT9	−3.8	271.9	345/105	70.0	17.2	5	4.9658	117	7.1	69.1	283.6
AT9-2	23.1	12.9	345/105	−31.2	3.3	8	7.8436	45	12.9	62.7	45.9
AT10	53.2	354.6	13/17	55.2	18.4	8	7.9435	124	5.0	75.2	343.5
AT11	46.4	3.2	13/17	46.6	21.3	10	9.8411	57	6.5	69.9	5.2
AT12	55.3	282.3	304/67	31.0	355.9	10	9.7601	38	8.0	69.6	85.9
AT13	40.3	312.1	340/25	47.4	335.7	7	6.8967	58	8.0	68.1	149.6
AT14	29.9	302.9	320/60	28.0	340.1	5	4.9295	57	9.9	61.9	118.7
AT15-2	57.3	4.6	150/75	30.9	271.2	6	5.9753	203	4.7	86.0	320.6
AT16-2	62.6	125.5	155/60	39.8	213.5	4	3.9667	90	9.7	4.3	290.0

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TABLE 2. (continued)

Site	Uncorrected			Corrected						Pole	
	I	D	Stk/Dp	I	D	N	R	k	α_{95}	Latitude	Longitude
Amalia Tuff (continued)											
AT17	57.9	34.9	230/50	26.0	354.8	6	5.9695	164	5.2	66.5	87.4
AT18	44.2	10.8	66/26	62.2	37.4	8	7.9569	162	4.4	61.0	320.0
AT19	55.3	15.5	328/27	32.9	30.8	14	13.9671	395	2.0	56.9	11.6
AT20	58.2	15.8	348/47	27.5	53.6	7	6.9963	1627	1.5	37.7	355.1
AT21	51.7	343.6	318/29	33.8	5.7	10	9.9045	94	5.0	71.1	57.7
AT22	50.6	1.2	171/30	46.5	326.1	10	9.9539	195	3.5	60.2	157.0
Avg	49.7	332.1		46.5	6.0	15	13.9999	14	10.6	80.7	38.2
Avg 2	46.2	20.0		−25.5	26.3	8	5.7195	7	38.1	33.8	44.0
Rhyolite Flows and Intrusives											
R1	74.0	292.0	2/40*	64.6	79.3	11	10.8104	53	6.3	32.5	308.1
R2	45.2	303.9	2/40*	68.1	1.6	8	7.9549	155	4.5	75.6	258.8
R3	8.9	325.6	67/40	47.9	320.1	10	9.9234	117	4.5	55.8	163.1
R4	−22.2	131.6	335/50*	−31.6	158.6	6	5.8488	33	11.8	−62.2	302.9
R5†	61.4	36.6				16	15.9511	307	2.1	61.5	321.7
R6	26.0	289.7	331/50*	47.4	327.1	7	6.9947	1125	1.8	61.2	157.4
R8	14.5	252.2	342/70*	84.5	254.3	8	7.9896	674	2.1	33.2	241.9
R9	32.3	305.5	2/40*	59.6	339.1	6	5.9849	332	3.7	73.3	183.5
R10	28.9	318.9	2/40*	49.0	349.0	5	4.9815	216	5.2	78.6	130.8
R11†	52.5	272.6				7	6.9541	131	9.8	21.0	190.8
Avg	33.2	301.1		60.9	342.6	8	7.3749	11	17.3	75.1	206.2
Santa Fe Group Sedimentary Rocks and Basalt Flows											
S1	−30.5	133.7	317/30	−27.6	150.7	6	5.8186	28	13.0	−55.8	311.7
S3	45.3	294.5	297/40	34.4	328.7	7	6.8042	31	11.1	57.2	139.9
AB1	22.3	352.0	10/20	27.0	1.1	5	4.9896	386	4.2	67.4	68.9
AB2	75.0	353.0	354/28	58.8	54.2	8	7.7071	24	11.6	48.0	323.8
SB1†	42.1	358.0				6	5.9299	71	8.0	77.2	83.0
Avg	45.7	325.4		40.8	351.0	4	3.5762	7	37.2	77.5	98.6

I, inclination in degrees; D, declination in degrees, uncorrected/corrected for structural tilt; Stk/Dp, strike and dip of structural attitude in left-hand azimuthal coordinates in degrees; N, number of samples averaged; R, vector sum; k, precision parameter; α_{95}, radius of 95% confidence in degrees (for sites with Bingham statistics: α_{95} is an average of the two Bingham 95% confidence limits, $k = k_1$ of the two concentration parameters, and R is calculated from k_1); latitude and longitude, north latitude and east longitude of the virtual geomagnetic poles in degrees; and Avg., unit mean of site-mean directions.

*Attitude taken from a nearby site

†Excluded from mean calculation.

within 1 km), and usually only a single correction was available for several sites (Table 2).

Alternating field demagnetization to peak fields of 100 mT on a majority of the quartz latite samples was successful in demagnetizing most of these rocks (J_{NRM} values are generally around 10^{-1} A m^{-1}), although thermal (above 600°C) and IRM experiments invariably showed the presence of hematite. At several sites (QL3, QL7, and QL8) a significant portion (20–60%) of the remanence is carried by hematite, and at site QL11, all of it is. The consistency of directions between titanomagnetite and hematite components in these rocks, as well as polished section observations, indicate that the hematite formed as a high-temperature oxidation product during emplacement and cooling of the quartz latites. Most quartz latite samples have a single stable component of magnetization upon demagnetization. However, in samples from sites QL5, QL7, and QL8, two separate stable components of magnetization are observed (Figure 4*c*). Secondary components that show directions near that of the predicted mid-Tertiary field before structural correction were usually removed in these rocks by 40 mT or 350°C, and the more resistant primary direction was generally isolated above 50–65 mT or between 400° and 610°C. J_s-T measurements, which define Curie temperatures between 560° and 585°C, and polished-section observations indicate predominantly titanomagnetite in most samples. The presence of maghemite is suggested from J_s-T curves at a number of sites (QL4, QL6, QL8, QL14, and QL16), and at sites QL8 and QL14, the presence of pyrrhotite is also inferred (Figure 4*c*).

Least squares lines were fitted to all components of magnetization, except at two sites (QL12 and QL13), where overlapping components were fitted by least squares planes. Although the average NRM intensities for these two sites were not distinctly higher than for the other quartz latites, overprinting by lightning-induced magnetizations remains a possibility, especially at these sites from higher elevations. Site-mean directions calculated for the component(s) of magnetization observed at each quartz latite site are plotted in Figure 6. Within-site dispersion for these mean directions is generally low (Table 2), and they basically form two distinct groupings (Figure 6*a*). Those site means with west-northwest declinations and shallow inclinations (before structural correction), which include only primary magnetizations from multicomponent samples, are interpreted to represent normal polarity magnetizations, acquired during the initial cooling of the quartz latites, that reflect the northeastward tilting of the units. The second group of site means, which includes only secondary magnetizations from multicomponent samples, has directions similar to the normal polarity predicted field for the region before structural correction. This predicted field direction in the quartz latites is considered to result from the partial remagnetization (or at sites QL1, QL2, and QL3 near the

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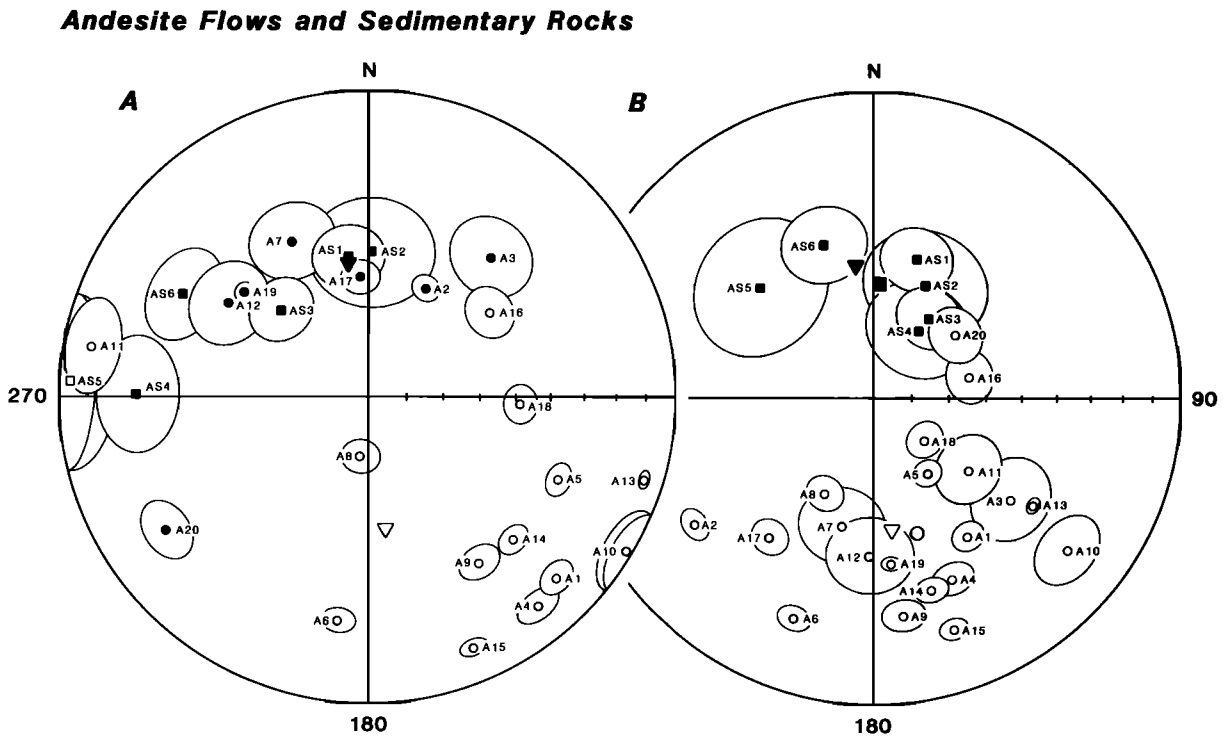


Fig. 5. Site-mean directions from the andesite flows (circles) and interbedded sedimentary rocks (squares; Table 2). Inverted triangle indicates predicted mid-Tertiary geomagnetic field direction at Questa (Table 3). (*a*) Directions uncorrected for bedding tilt and (*b*) corrected directions. In Figure 5*b*, normal polarity andesite flows have been reversed, so that all andesites are of reversed and all sedimentary rocks are of normal polarity. Large symbols indicate averaged directions (Table 2). Diagrams are equal-area plots, where open symbols indicate projection from the upper hemisphere, and closed symbols from the lower hemisphere.

south caldera margin, the total remagnetization) of these rocks after the regional structural deformation had ended. However, at site QL8 the secondary component could be carried by maghemite (Figure 4*c*), having been acquired under recent geomagnetic field conditions (Table 3). Site means QL7 and QL8 have primary directions that were acquired under reversed field conditions prior to structural tilting.

A comparison of the **k** values associated with the 13 mean directions for those sites exhibiting predeformation magnetizations (QL1–QL3 rejected) shows an improvement from 4 before to 11 after the application of the structural correction. Although the corrected directions move toward the predicted mid-Tertiary direction and the precision is increased, the test is not significant at the 95% confidence level, most likely due to the small number (six) of available corrections. For many of the small intrusive bodies of quartz latite, only their directions of primary magnetization demonstrate the structural tilting of these units.

Amalia Tuff and Other Rhyolitic Rocks

The 26-m.y.-old Amalia Tuff was widely sampled at 17 sites in erosional remnants of the regional ash flow sheet outside the Questa caldera (Figure 3), including two sites (AT15 and AT16) in the Brazos-Tusas Mountains to the west of the San Luis Valley (Figure 1). Five sites were also collected in the thick intracaldera accumulation of welded tuff preserved along the southern margin of the caldera. Typically, the outflow sheet is only a few tens of meters thick, and likely cooled within a short period (1–10 years) of time relative to the secular variation of the earth's field (1000–10,000 years). Furthermore, a structural correction was taken at each site from the eutaxitic structure of the tuff (a planar structure formed from collapse and compaction of pumice fragments within the ash flow during welding [e.g., *Ross and Smith*, 1961]), allowing an evaluation of this inferred paleohorizontal indicator. Samples were usually taken in the densely welded, devitrified portion of the ash flow sheet, but at sites AT2, AT10, and AT19, samples were collected from the partially welded vapor phase zone as well.

The rhyolitic rocks (sampled at 11 sites) are presented as a group on the basis of broadly similar compositions and magnetic properties, although some differ widely in age, diverging from the stratigraphic order of this presentation. Sampled in flow-banded lavas or small intrusions and dikes, many of

TABLE 3. Predicted Mid-Tertiary and Present-Day Field Directions at Questa

							Pole	
	I	*D*	*N*	*R*	*K*	A_{95}	Latitude	Longitude
Predicted	54.0	352.1	102	94.1021	13	4.1	83.2	148.0
Present-day	64.7	12.8					76.3	294.5
Dipole field	56.2	0.0					90.0	0.0

See Table 2 footnote for explanation of variables. Predicted, average direction calculated from mid-Tertiary paleomagnetic poles for North America [*Diehl et al.*, 1983]. Present-day, present-day geomagnetic field direction. Dipole field, present-day direction calculated assuming an axial geocentric dipole.

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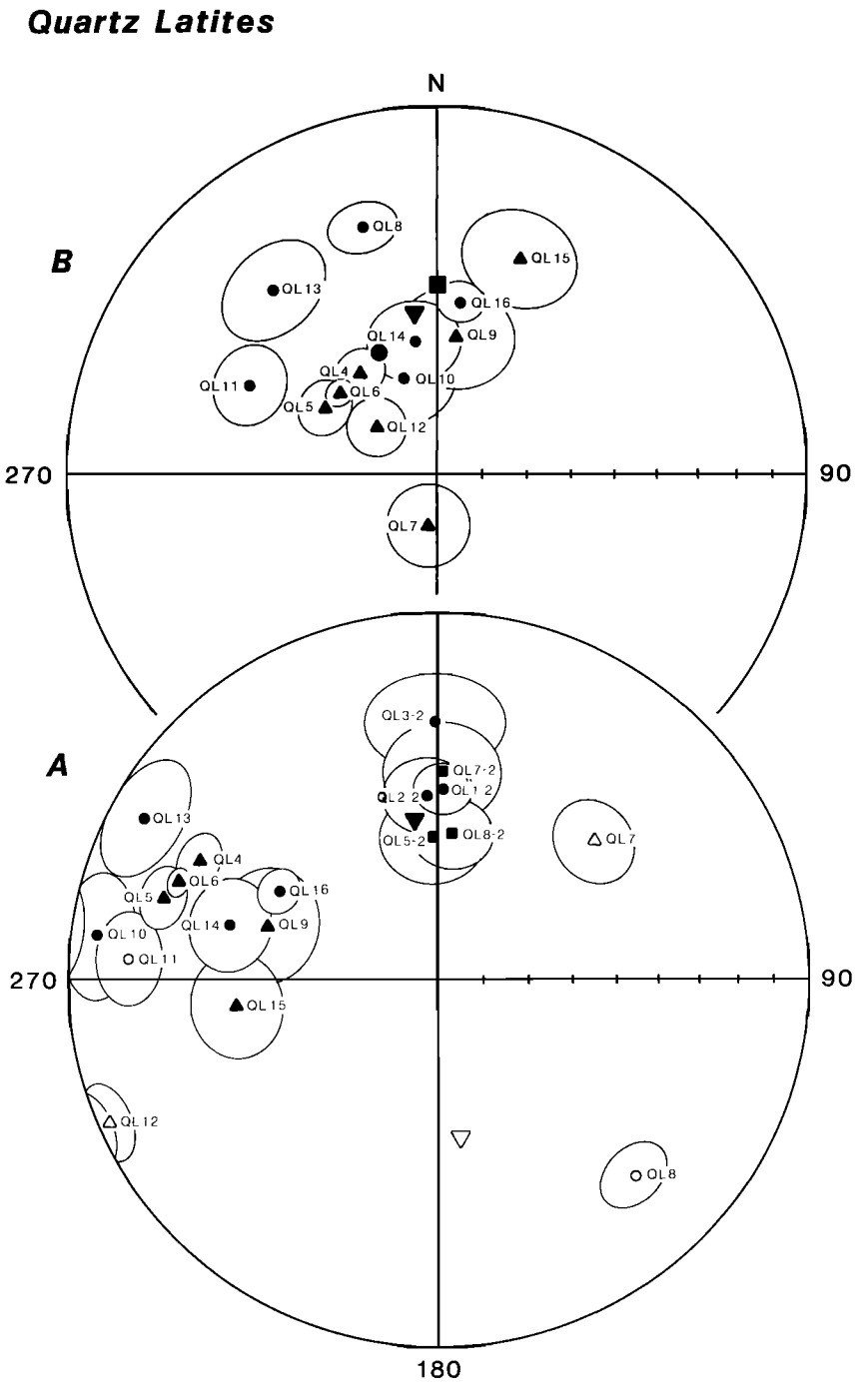


Fig. 6. Site-mean directions from the quartz latitic rocks (Table 2). (*a*) Uncorrected directions and (*b*) corrected directions with reversed polarity directions switched to normal polarity ones for comparison. Secondary directions (−2) are indicated by squares in Figure 6*a*. Inverted triangle indicates predicted mid-Tertiary geomagnetic field direction at Questa (Table 3). Large symbols in Figure 6*b* represent averages of the corrected primary directions (circle) and the uncorrected secondary directions (square).

these rocks are closely associated with the eruption of the Amalia Tuff. However, site R6 was taken in a domical body related to the weakly indurated rhyolitic tuff of Tetilla Peak (site R7), which predates the andesite flows [*Lipman*, 1983]. Sites R9 and R10 were collected from the comenditic rhyolite lavas that immediately preceded the eruption of the Amalia Tuff, and at site R5, a 7- to 11-m.y.-old rhyolite dome (Table 1) was sampled.

The samples of both the Amalia Tuff and rhyolitic rocks exhibit, with few exceptions, single-component magnetizations upon thermal and alternating field demagnetization. AF treatment to 100 mT generally reduces the J_{NRM} (usually 10^{-1}–10^{-2} A m^{-1}) to less than 10–15% of the initial values (Figure 4*d*). Thermal and IRM experiments indicate that the primary remanence carrier is titanomagnetite, with minor amounts of similarly magnetized authigenic hematite present. At sites AT6 and AT19 (vapor phase zone), a significant portion of the remanence is carried by hematite, and at AT10, also in the vapor phase zone, hematite is the dominant carrier. Thermomagnetic determinations yield Curie temperatures in the range 547°–601°C, although lower values (486°–508°C) are found in the intrusive rhyolites and suggest that the magnetic

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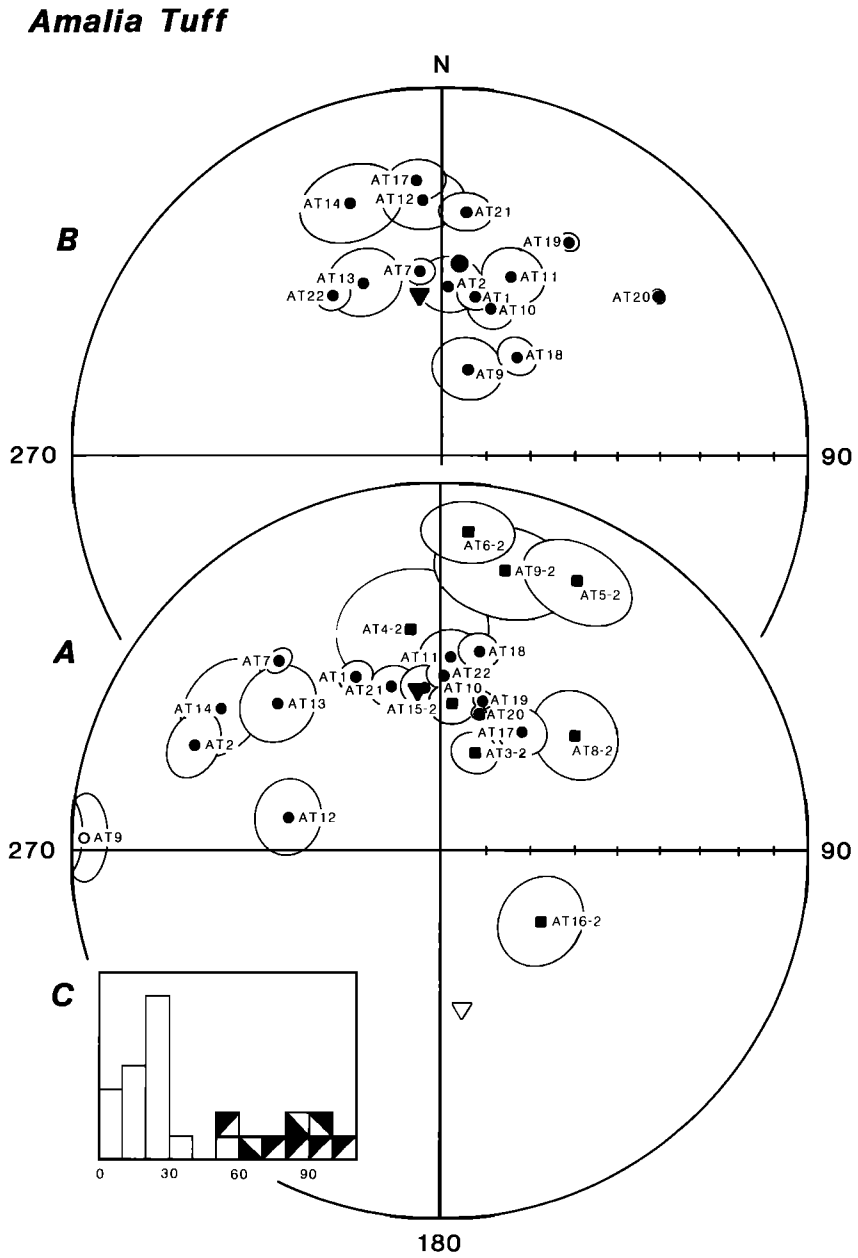


Fig. 7. Site-mean directions from the Amalia Tuff (Table 2), (*a*) uncorrected and (*b*) corrected. Squares in Figure 7*a* represents remagnetized sites (−2; see text) that have been excluded from Figure 7*b*.) (*c*) Histogram of number of sites against angular distance from the unit mean (large solid circle in Figure 7*b*). Inverted triangle indicates predicted mid-Tertiary geomagnetic field direction at Questa (Table 3). Semisolid boxes indicate discarded sites: solid upper left, sites from the intracaldera fill near the southern caldera margin; solid lower left, sites from the Brazos-Tusas Mountains; solid upper right, site AT8 near the Rito del Medio pluton.

mineralogy in these rocks ranges from titanomagnetite to magnetite and ilmenohematite to hematite. At several sites in the tuff and rhyolites, the J_s-T results are also indicative of small amounts of maghemite and pyrrhotite. Spurious magnetizations were removed first in demagnetization from many samples but rarely persisted above 30 mT or 350°C. However, at site R3, the linear decay of the vector end points under AF demagnetization was to a point away from the origin, suggesting that a resistant component of magnetization, probably associated with hematite, was acquired at a time differing from that of acquisition for the removed component. High NRM intensities ($J_{NRM} > 10$ Am^{-1}) at sites AT14, R9, and R10 suggest lightning-related effects. Results from sites AT23 and AT24, also with high intensities and collected on a ridge crest near recent strikes, were discarded due to dispersed magnetizations that are almost certainly lightning induced. Site R7 was also discarded on the basis of highly dispersed magnetizations that are likely associated with the abundant lithic fragments contained in this older, poorly indurated nonwelded tuff [*Geissman*, 1980, p. 1494].

Demagnetization planes were needed to resolve magnetic components in the tuff and rhyolitic rocks at three sites. At sites AT14 and R11, the overprinting is inferred to be lightning related, although at R11, abnormally high J_{NRM} values were not observed. The overprint component observed at site AT9, interpreted to be associated with later oxidation and resolved with demagnetization planes, resides in portions of the coercivity and blocking temperature spectra higher (hema-

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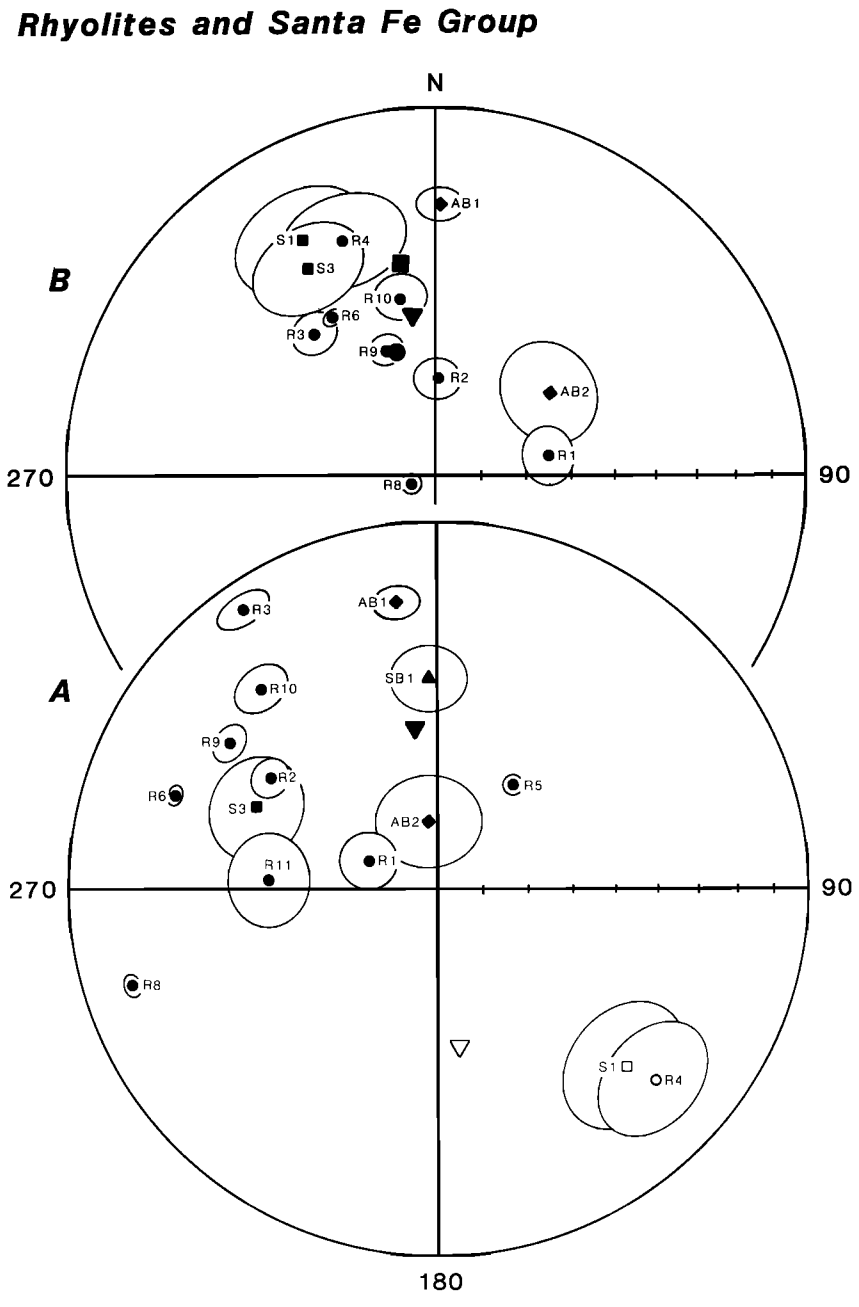


Fig. 8. Site-mean directions from rhyolitic rocks (circles), late rift-related sediments (squares), and interbedded basalt flows near Amalia (diamonds), and the site from the Servilleta Basalt (triangle; Table 2). Inverted triangle indicates predicted mid-Tertiary geomagnetic field direction at Questa (Table 3). (*a*) Uncorrected directions and (*b*) corrected directions with reversed polarity sites switched to normal polarity. Large symbols indicate averaged directions (Table 2).

tite) than the primary component (titanomagnetite), and is identified as a secondary component because it moves away from the predicted field direction upon correction.

Figure 7 shows the site-mean directions for the Amalia Tuff before and after the structural corrections have been made using the eutaxitic structure. The directions for most site means after correction ($N = 15$) converge on the predicted mid-Tertiary direction with a reduced dispersion that is significant at the 95% level, demonstrating that the magnetization predates the deformation and that the eutaxitic structure is a relatively reliable indicator of the paleohorizontal. However, at eight sites, mean directions were rotated from a position near the predicted direction to one well away from it, suggesting remagnetization at these sites postdating the deformation. These sites have been withdrawn from the after-correction diagram. A histogram of corrected site means (Figure 7*c*), comparing their angular distance from the unit-mean direction, shows the two populations. All sites in the thick intracaldera fill near the mineralized granitic intrusions along the caldera's southern margin have a remagnetized component consistent with other rock units in this area. Only at site AT9 is a predeformational component found. The other sites included in this remagnetized group are the two sites (AT15 and AT16) from the western margin of the rift valley and a site (AT8) along the eastern margin near the Rito del Medio pluton. Site AT20, which may be remagnetized (Figure 7), has not been rejected from the corrected unit mean because it is far from any younger intrusive rocks. Another possibility for

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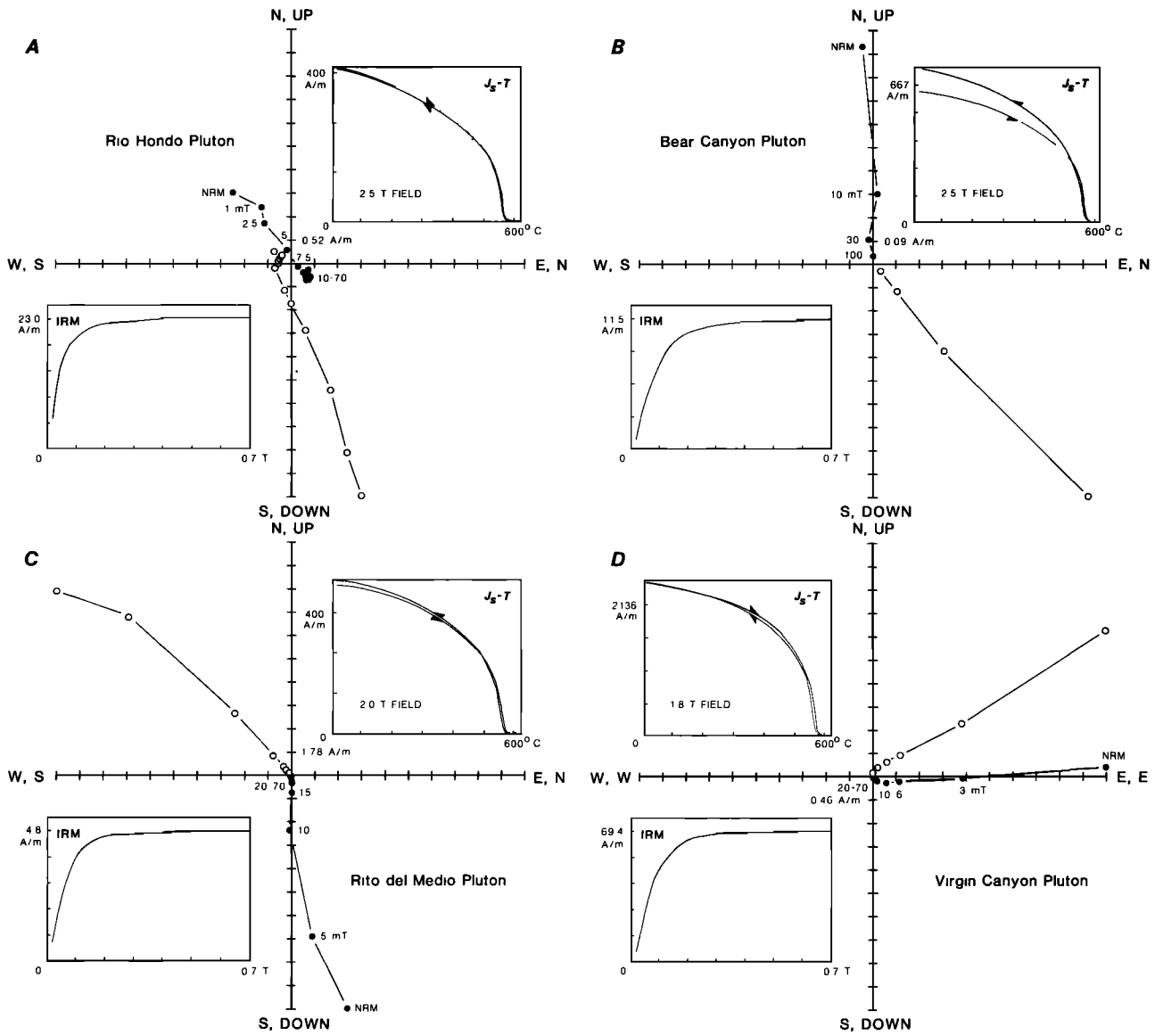


Fig. 9. Orthogonal projections (uncorrected for bedding tilt) of demagnetization sequences for samples from the intrusive units associated with the Latir volcanic field. (*a*) AF results from the Rio Hondo pluton in Columbine Creek at site RH12; insets show J_s-T and IRM results on specimens from the same site. (*b*) AF results from the Bear Canyon pluton at site BC1. (*c*) AF results from the Rito del Medio pluton at site RM3. (*d*) AF results from the Virgin Canyon pluton at site VC3. Solid circles, projections of vector end points onto horizontal plane; open circles, onto vertical plane.

the divergent group of directions is that the eutaxitic structure measured in these rocks does not accurately reflect the paleohorizontal at the time of initial magnetization.

Uncorrected site-mean directions for the rhyolites (Figure 8*a*) are also mostly of normal polarity and consistent with the established regional structural deformation. Only site R4, from Relica Peak to the south, has a tilted reversed polarity direction, and the site direction (R5) from the young dome near Amalia appears untilted. Structural corrections for some intrusive rhyolites are unavailable (R5 and R11), and for the lavas, the flow banding proves to be an unreliable measure. General corrections for eastward tilts, supplied by nearby units, yield results closer to the predicted field direction (Figure 8*b*, Table 2).

Rift-Related Sedimentary Rocks and Associated Basalts

Widely overlying the volcanic rocks of the Latir field, the weakly indurated clastic sedimentary sequences of the Santa Fe Group were sampled at three sites in the northern portion of the study area (S1–S3, Figure 3). These rocks dip as much as 45°E, although the dips tend to decrease upsection. Alkalic basalt flows (15–16 m.y. old), locally interlensing with the sedimentary rocks and dipping up to 30°E, were sampled at two sites, and the flat lying 4-m.y.-old Servilleta Basalt, capping the entire stratigraphy, was sampled at one site.

Two sites from the finer grained sedimentary rocks (S1 and S3) have similar demagnetization and IRM results that point to hematite as the primary remanence carrier. Heatings to 670°C were employed on most samples, yielding single-component magnetizations above 400°C. Coarser grained samples collected at site S2, however, have erratic AF and thermal demagnetization behavior and are discarded from further analysis. AF treatment to 100 mT proved sufficient to demagnetize the alkalic basalt and Servilleta Basalt, which have a single stable component, mostly above 20–30 mT. At site AB1, spurious overprints remained in many samples to high demagnetization levels, requiring the use of demagnetiza-

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tion planes to provide the most accurate site-mean direction. A Curie temperature of 482°C was determined for the alkalic basalts, and one of 577°C, indicating nearly pure Fe_3O_4, for the tholeiitic Servilleta Basalt.

An average of the site-mean directions (Figure 8*b*) for the sediments and enclosed basalts is closer to the predicted field direction after structural correction. However, due to similar site corrections, no corresponding improvement in the site-level directional dipersion is seen. The large difference in direction between the two sites in the alkalic basalts is likely due to secular variation of the field between emplacement of the two flows. The direction obtained from the Servilleta Basalt near Amalia, New Mexico, is coincident with the result obtained by *Ozima et al.* [1967] from the Rio Grande gorge in the San Luis Valley, suggesting little relative tilting between these two locations since emplacement.

Southern Plutons

Two intrusive bodies were sampled in the more deeply exposed region to the south of the Questa caldera: the large 25- to 26-m.y.-old mesozonal instrusion exposed along Rio Hondo, and the younger (21–22 m.y. old) high-silica granite exposed on Lucero Peak (Figure 3). Six sites were collected in the quartz monzonite and two sites (RH3 and RH8) in the NW trending rhyolite dikes cutting the pluton in Hondo Canyon. At site RH2 in the quartz monzonite, samples were also taken from a large mafic inclusion (1–2 m in diameter). One site (RH10) was in an andesitic dike and a mafic inclusion (< 1 m in diameter) exposed in Italianos Creek, and exposures of the quartz monzonite (RH12 and RH9) and the higher level granite (RH11) were also sampled in the Columbine Creek drainage to the north. The Lucero Peak pluton was sampled at five sites.

Overall, the magnetic characteristics of the Rio Hondo pluton are uniform, although the results of demagnetization differ between the samples collected in Hondo Canyon and those from the Columbine Creek drainage. $\mathbf{J}_{NRM}$ values for all rock types are around 10^{-1} A m^{-1}, and AF demagnetization of the quartz monzonite and granite usually reduces the intensities to low levels (< 10%) within 30 mT. Higher coercivities (to 100 mT) are encountered in the finer grained rhyolitic dike and mafic inclusion rocks. Thermal demagnetization to 610°C, AF results, and IRM and $\mathbf{J}_s - \mathbf{T}$ (T_c = 575–582°C) measurements on all rocks as well as polished section observations show that the magnetic carrier is pure or nearly pure magnetite.

Demagnetization results from Hondo Canyon are generally univectorial and of normal polarity with two exceptions: the mafic inclusion sampled at site RH2 shows multivectorial behavior defining a single consistent component above 20 mT. The high-$\mathbf{H}_c$ component recovered between 20 and 100 mT from three samples in the inclusion is of opposite polarity to that exhibited by five samples in the enclosing quartz monzonite (Figure 10*a*). Three samples taken from the other mafic inclusion in Italianos Creek gave demagnetization planes that converge on an atypical reversed polarity direction and were not included in the RH10 site average. From the Columbine Creek drainage, however, the results are mostly multivectorial (Figure 9*a*). A mean direction of reversed polarity for sites RH9 was calculated from demagnetization planes. At site RH12, demagnetization lines were fitted to the overprint component, and the primary component is defined by a combination of demagnetization lines and planes [*Kirschvink*, 1980, method 2] At site RH11, lines were fitted to those portions of the demagnetization curves representing a reversed polarity component.

The Lucero Peak pluton gives results that vary in quality but are generally consistent in direction. A single, normal polarity component of magnetization is removed from these rocks, commonly by 50 mT ($\mathbf{J}_{NRM} = 10^{-2}$ Am^{-1}), that is probably carried by pure magnetite (T_c = 582°C). For site LP1, **k** is high (174), indicating a consistent magnetization. The other sites yield **k**'s between 10 and 35, and for sites LP5, **k** = 8 (Table 4). Three samples were discarded from site LP5 showing directions that move to scattered negative inclinations upon demagnetization. The range of dispersion in these rocks is perhaps related to the cooling rate of the pluton at each site. With a faster cooling rate, a shorter period of secular variation would be recorded, and the expected within-site dispersion would be less. Thus the cooling period is interpreted to have been short relative to secular variation at site LP1, and at site LP5, long enough to have also included a polarity reversal. The general agreement of the majority of the site means from the Lucero Peak granite as well (Table 4) suggests that the secular variation has been averaged.

The similarity in direction between the magnetizations of the Rio Hondo and Lucero Peak plutons and the predicted field suggests that the emplacement and cooling of both plutons postdated the regional deformation and is consistent with their subhorizontal roof contacts [*Lipman*, 1983].

Plutons Along the Southern Caldera Margin

The three 22- to 23-m.y.-old postcaldera granitic plutons exposed by the Red River along the southern caldera margin were sampled at 16 sites (Figure 3). Five sites were taken in the Sulfur Gulch pluton (including one from a cross-cutting aplite dike), two from the Bear Canyon pluton, and nine from the more compositionally and texturally complex intrusion near Red River, New Mexico. The sites collected from the complex near Red River include granite and rhyolite porphyry (RR1, RR7, and RR8), aplite (RR2), quart monzonite porphyry (RR3), quartz latite (RR4 and RR5), and fine-grained monzonite (RR9). A rhyolitic dike from a large east-west swarm associated with the intrusion near Red River was also sampled (RR6).

The Sulfur Gulch and Bear Canyon plutons yield results similar to those from the Lucero Peak pluton and show a single, normal polarity component of magnetization near the predicted field direction (Figure 10*b*). Coercivities for this component are generally less than 30–50 mT in the Sulfur Gulch pluton but higher (up to 100 mT) for samples from the cross-cutting aplite dike (SG5) and the Bear Canyon pluton (Figure 9*b*). $\mathbf{J}_{NRM}$ values are uniform for both units (10^{-2} A m^{-1}), and IRM and $\mathbf{J}_s$-**T** procedures and polished section observations show the presence of primarily pure magnetite (T_c = 579°C) accompanied by some hematite. Within-site dispersion for the sites in the Sulfur Gulch pluton is large (Table 4), and one site (SG4), the only one taken from the underground workings and visibly pyritized, was discarded from the unit mean due to erratic demagnetization behavior. However, within-site dispersion for the Bear Canyon pluton is small (Table 4).

The intrusion at Red River gives a wider range of paleomagnetic results, in keeping with its more complex geology. $\mathbf{J}_{NRM}$ values vary with composition, giving higher values (10^{-1} A m^{-1}) for the mesocratic rocks than for the more silica-rich rocks (10^{-2} A m^{-1}). Most samples show multicomponent behavior upon demagnetization. At four sites (RR1, RR4, RR6,

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TABLE 4. Site-Mean Directions and Poles for the Intrusive Units

Site	Uncorrected						Pole	
	I	D	N	R	k	α_{95}	Latitude	Longitude
Southern Plutons: Rio Hondo and Lucero Peak								
RH1	51.1	323.1	10	9.4610	17	12.2	59.3	166.5
RH2	54.8	320.5	5	4.9337	60	9.9	58.2	174.1
RH2-R	−58.4	128.6	3	2.9931	319	6.9	−49.9	4.3
RH3	54.6	329.0	6	5.9067	54	9.2	65.0	170.4
RH4	57.9	343.7	8	7.5837	17	13.9	76.9	178.3
RH5	52.8	346.8	8	7.7427	27	10.8	78.7	151.9
RH6	60.3	352.4	7	6.6814	19	14.3	82.4	205.1
RH7	61.6	345.0	7	6.8905	55	8.2	76.9	197.4
RH8	52.4	337.4	6	5.9433	88	7.2	71.1	160.4
RH9-R	−23.8	136.1	5	4.9803	203	4.8	−43.9	324.4
RH10	65.7	343.9	12	11.5555	25	8.9	73.7	213.2
RH11-R	−34.5	145.1	4	3.9510	61	11.8	−54.7	324.0
RH12	77.3	43.9	14	13.6342	36	6.8	51.5	281.1
RH12-R	−36.7	138.5	10	9.6419	25	13.8	−50.5	331.9
Avg	54.3	331.3	14	13.4240	23	8.6	66.7	168.7
LP1	56.6	53.9	10	9.9483	174	3.7	47.4	326.8
LP2	53.9	12.3	7	6.8124	32	10.8	79.8	352.4
LP3	57.4	358.7	5	4.7125	14	21.3	88.2	219.4
LP4	58.7	358.0	7	6.6672	18	14.6	72.4	180.5
LP5	50.3	6.5	5	4.5163	8	28.3	82.3	28.3
Avg	57.8	9.7	5	4.8579	28	14.7	82.2	326.5
Southern Caldera Margin Plutons: Sulfur Gulch, Bear Canyon, and Red River								
SG1	42.2	338.7	7	6.6295	16	15.5	68.0	136.5
SG2	57.3	8.3	7	6.2841	8	22.1	83.3	331.9
SG3	51.3	353.1	6	5.7576	21	15.1	82.6	126.8
SG5	57.9	356.9	5	4.6002	10	25.4	86.9	203.0
Avg	52.7	352.9	4	3.9493	59	12.0	84.0	136.1
BC1	48.8	4.2	8	7.9654	202	3.9	82.2	46.8
BC2	56.0	357.0	11	10.9246	133	4.0	87.6	161.3
Avg	52.4	0.9	2	1.9946	184	18.5	86.4	64.8
RR1-R	−42.4	114.0	6	5.9481	96	8.3	−33.0	352.3
RR1	53.8	353.3	7	6.7840	28	11.6	84.1	143.0
RR3	39.1	36.5	7	6.6840	19	14.2	55.3	359.1
RR4	53.2	346.8	7	6.5908	15	16.3	78.8	153.0
RR4*	−0.9	220.1	6	5.6011	13	19.7	−38.2	199.6
RR5	45.2	359.8	6	5.6359	14	18.7	80.0	75.6
RR6	61.4	346.6	6	5.8443	32	12.0	78.2	198.2
RR6*	−22.5	207.3	4	3.9268	41	14.5	−55.0	203.1
RR7-R	−43.9	135.6	4	3.8666	22	19.8	−50.9	341.2
RR7	56.4	22.6	6	5.7660	21	14.8	72.0	337.0
RR8-R	−47.6	120.5	4	3.9958	720	3.0	−40.1	353.6
RR9	−49.8	196.8	5	4.9217	51	10.8	−74.7	183.5
Avg	54.5	346.9	10	9.2590	12	14.4	79.3	158.9
Intracaldera Plutons: Cabresto Lake, Rito del Medio, Virgin Canyon, and Cañada Pinabete								
CL1	50.0	15.7	7	6.9445	108	5.8	75.7	4.5
CL2	40.0	10.1	4	3.9125	34	15.9	73.6	39.8
CL3	56.9	31.9	6	5.6360	14	18.7	64.7	333.0
Avg	49.4	18.0	3	2.9625	53	17.0	73.4	0.4
RM1	44.7	346.0	9	8.3839	13	14.9	74.1	127.1
RM2	44.9	9.5	6	5.5694	12	20.5	76.9	33.5
RM3	−44.5	163.0	4	3.9176	36	15.4	−72.1	313.1
RM4	−21.3	181.6	6	5.7721	22	16.2	−64.2	250.8
Avg	39.4	355.5	4	3.8965	29	17.4	75.5	92.5
VC1-2	−44.4	131.3	4	3.9904	314	5.2	−47.5	344.3
VC2	−19.2	77.9	4	3.9918	365	4.8	3.6	359.6
VC2-2	65.5	325.1	5	4.9733	150	6.3	62.2	198.7
VC3	−30.5	84.8	4	3.9708	103	8.0	−5.7	0.8
VC3-2	−53.1	133.5	4	3.9719	107	8.9	−52.3	353.9
VC4*	−37.4	169.0	5	4.9016	41	12.1	−71.5	169.0
VC5*	62.5	74.8	7	6.8057	31	11.0	34.5	312.0
Avg	24.9	261.2	2	1.9874	79	28.5	1.0	180.2
Avg 2	54.5	315.4	3	2.9611	51	17.4	54.8	176.7
CP1	73.2	61.7	7	6.5624	14	16.9	45.1	294.6

See Table 2 footnote for explanation of variables.
*Excluded from mean calculation.

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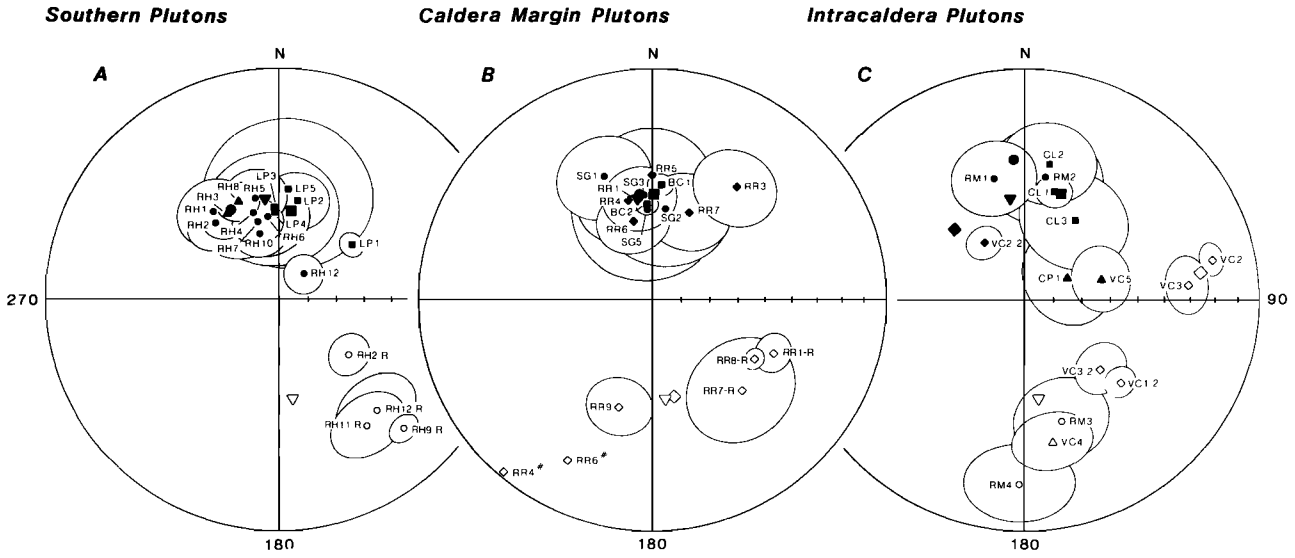


Fig. 10. Site-mean directions from the intrusive rocks (Table 4). Inverted triangle indicates predicted mid-Tertiary geomagnetic field direction at Questa (Table 3). Larger symbols represent averages of site directions (see text). (*a*) Southern plutons: squares, Lucero Peak pluton; circles, Rio Hondo pluton; and triangles, NW trending rhyolitic dikes cutting the Rio Hondo pluton. (*b*) South caldera margin plutons: circles, Sulfur Gulch pluton; squares, Bear Canyon pluton; and diamonds, Red River intrusive rocks (unit average of reversed polarity for clarity). (*c*) Intracaldera resurgent plutons: circles, Rito del Medio pluton; squares, Cabresto Lake pluton; diamonds, Virgin Canyon pluton; and triangles, Virgin Canyon south wall intrusives and Cañada Pinabete pluton (CP1).

and RR7) a consistent normal polarity secondary component is observed. The secondary components are usually removed by 30 mT or 400°C, but at sites RR1, RR7, and RR8 the reversed polarity high-coercivity (>100 mT) portions of the remanence were isolated by heating to temperatures greater then 600°C, suggesting a hematite carrier. IRM and $\mathbf{J}_s$-T measurements, however, show that magnetite ($\mathbf{T}_c$ = 574–582°C) is the predominant magnetic mineral in these rocks. The recovery of the reversed polarity direction for sites RR4 and RR6 is thought to be impaired by the incomplete removal of the normal polarity direction during demagnetization, and these sites are excluded from the unit-mean calculation. Sample results from one site, a gray colored pyritized aplite (RR2), were not included in overall averages due to unstable demagnetization behavior.

The unit-mean directions from the Sulfur Gulch, Bear Canyon, and Red River plutons indicate no significant tilting of these rocks since their magnetization, although the possibility remains that the potentially pretilting portions of the Red River intrusive complex have been overprinted by younger untilted portions of the complex.

Central and Northern Intracaldera Plutons

The 25- to 26-m.y.-old resurgent granitic plutons within the central and northern parts of the caldera were sampled, including the Cabresto Lake pluton at three sites, the Rito del Medio pluton at four sites, the Cañada Pinabete pluton at two sites, and the Virgin Canyon pluton at five sites (Figure 3).

From the Cabresto Lake and Rito del Medio plutons, mostly normal polarity, single-component magnetizations were removed during demagnetization, usually by 30 mT. However, site RM3 in the Rito del Medio pluton shows a similar but reversed polarity component (Figure 9*c*), and at site RM4, both the normal and reversed polarity components are found. $\mathbf{J}_s$-T and IRM experiments and polished section observations show mainly magnetite (T_c = 570–581°C) in these rocks, with subordinate amounts of secondary hematite present in the Virgin Canyon and Rito del Medio plutons. A greater than average amount of hematite is present at site RM4 and appears to carry a significant portion of the high-coercivity (>100 mT) reversed polarity component, isolated with demagnetization planes. The Cañada Pinabete pluton is unusual in that it shows $\mathbf{J}_{NRM}$ values of 10^{-4} A m^{-1} compared with around 10^{-1} A m^{-1} for the other intracaldera plutons. Magnetization intensities approached the lower limit of the magnetometer's resolution during demagnetization of the Cañada Pinabete samples. At one site from the fine-grained peralkaline border phase (CP2), results were discarded due to unacceptably noisy demagnetization curves. AF and thermal demagnetizations were successful in unraveling the multivectorial behavior in most sites from the Virgin Canyon pluton. Hematite accompanies the magnetite (T_c = 578°C) in these rocks and is associated with the higher coercivity component at site VC2 (Figure 9*d*). The two sites from the southern portion of the Virgin Canyon pluton in granites of metaluminous (VC4) and peralkaline (VC5) compositions give equivalent results to those that characterize the Rito del Medio pluton and are excluded from the unit-mean calculation for the Virgin Canyon pluton. Site means for the intracaldera plutons are given in Table 4.

The site-mean directions for the intracaldera plutons (Figure 10*c*) are near the normal or reversed polarity predicted field direction for the region, indicating that most of these rocks became magnetized after the regional deformation had waned. The Virgin Canyon pluton, however, exhibits magnetizations that predate the eastward tilting of the area. Such a tilt (70°) is consistent with its peralkaline composition, which potentially makes it an intrusive equivalent of the Amalia Tuff [*Lipman*, 1983]. Tilted reversed polarity directions associated with coercivities and blocking temperatures indicative of magnetite are found at sites VC2 and VC3, each showing a relatively untilted secondary component of higher coercivity (Figure 10*c*). These two secondary directions are essentially the same in direction and of opposing polarity; both polarities

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of this direction are found at site VC1. This site (VC1) in the Virgin Canyon pluton, however, became magnetized after most of the structural tilting had occurred. A site from the andesite flows structurally above the pluton in Virgin Canyon (A18) shows a similar reversed polarity, strongly tilted direction (Figure 5a).

DISCUSSION

Secular Variation

For most of the plutonic rock units and volcanic sequences sampled in this study, the interval of time spanned by their cooling and subsequent magnetization is likely greater than that required (approximately 10^4–10^5 years) to average out the secular variation (SV) of the geomagnetic field [*Merrill and McElhinny*, 1983]. However, for a single site or unit with samples limited to a time interval insufficient to completely average SV, the distinction between dispersion due to SV and that due to structural deformation is important.

Model values for the angular standard deviation of directions (σ) for the latitude of northern New Mexico are approximately 20° [*Cox*, 1970; *McFadden*, 1980]. Probably the best estimate of the contemporaneous SV for this study comes from a paleomagnetic investigation [*Beck et al.*, 1977] of the relatively undisturbed similarly aged rocks of the San Juan field (Figure 1) that yields $\mathbf{s} = 21.8°$, recalculated for the Latir field latitude [*Cox*, 1970]. Since 95% of the sample population is expected to fall within 2σ of the mean direction, or roughly 40°, a site-mean direction that is more than 40° from the predicted direction might reflect a component of structural deformation. This 40° limit is directly applicable to individual sites in the volcanic units, which cooled quickly and may reflect the full range of SV, although for sites in the slower cooling plutonic rocks, which tend to average some portion of the SV, this limit represents an upper bound.

The single ash flow sheet of the Amalia Tuff cooled and became magnetized over a very short period of time (<20 years) relative to SV [*Riehle*, 1973] and likely provides an accurate record of the ambient field direction. However, the after-correction site-mean dispersion remains high ($\mathbf{s} = 21.7°$). This dispersion is due to errors in the structural correction, based on the tuff's eutaxitic structure, and in part to errors in the site means. An estimate of the site-mean errors using the site α_{95} values shows them to be negligible, and assuming errors in the field measurements are also negligible, a mean angular deviation of 18.7° is calculated. Assuming block rotations about vertical axes are insignificant, this mean deviation can be considered an approximate estimate of error for the dip of the eutaxitic structure in its representation of the paleohorizontal. This error can be accounted for either by an initial dip of the eutaxitic structure or by structural disruption of the tuff between the time of compaction and the time of magnetization. For comparison, the mean angular deviation for 33 sites in the Wah Wah Springs Tuff, also a single cooling unit, is 13.8° [*Shuey et al.*, 1976].

Inferred Structural History of the Latir Volcanic Field and Relations to Regional Episodes of Extension

The paleomagnetic data, in conjunction with the available isotopic dates, provide temporal information important to constructing a detailed structural history of the Questa caldera and Latir volcanic field. In a number of cases, the paleomagnetic results resolve ages for rocks, relative to tilting, within the uncertainties of the isotopic dates. The marine magnetic anomaly record [e.g., *Lowrie and Alvarez*, 1981] indicates that the geomagnetic field reversed in polarity frequently during the period of time (Table 1) in which the rocks of the Latir field were emplaced, making correlations between units based on paleomagnetic polarity data difficult. However, the polarity data are useful in estimating durations for some igneous events.

Consistent with the isotopic dates, roughly equal numbers of andesite sites have normal and reversed polarity directions, suggesting these rocks were erupted over a period of time including both field polarities. Normal polarity directions for all of the sites in the interbedded sedimentary rocks are considered a coincidence. The consistency of the primary normal polarity directions for the quartz latite flows and intrusives (Figure 6) suggests that these rocks were emplaced in a relatively short period of time, primarily within a single polarity interval (<1 m.y.). Similar initial Sr isotopic ratios among the quartz latites (C. Johnson, unpublished data, 1984) supports the rapid eruption of these rocks from a single uniform source. However, the possibility that these rocks were magnetized over several polarity intervals cannot be conclusively ruled out. Two sites (QL7 and QL8) from the quartz latitic hypabyssal intrusive rocks and one site (R4) from rhyolitic rocks to the south of the caldera structure have tilted but reversed polarity directions, indicating they were magnetized at a different time. A normal polarity quartz latite dike (QL9) cuts the reversed polarity rocks and may be related to the subsequent widespread eruption of quartz latites in the Latir field.

The 26-m.y.-old Amalia Tuff erupted during a period of normal polarity. Collapse of the caldera structure was coincident with the removal of the large quantity of magma (>400 km^3) from a high level in the crust, and the tuff filled the resulting depression to a depth of 2 km or more [*Lipman*, 1983]. A reversed polarity sample from site AT9 and other reversed polarity sites within the thick intracaldera welded tuff [*Snyder*, 1984] indicate that these rocks cooled over an extended period of time (>1000 years [*Riehle*, 1973]) and became magnetized during a subsequent reversed polarity epoch. Similar maximum stratal tilts for the entire volcanic sequence suggest that tectonism began after eruption of the Amalia Tuff and was most intense within the caldera structure where the brittle crust was presumably thinnest. Apparently, this zone of brittle crust included the quenched upper portion of the magma chamber represented by the Virgin Canyon pluton, which cooled and was magnetized along with the intracaldera fill under reversed polarity field conditions before being tilted. Tilting (>40°), however, was not restricted to the caldera region but affected units extending from near Amalia in the north to the southernmost exposures of the Latir volcanic rocks (R4, QL8–QL9, Figure 3).

This pronounced episode of extension was short lived. Reversed polarity peralkaline rocks from the southern portion of the Virgin Canyon pluton and the Rito del Medio and Cabresto Lake plutons, both dated around 25–26 m.y. ago (Table 1), are not significantly tilted. In addition, high-coercivity, normal and reversed polarity components in the Virgin Canyon pluton, probably related to deuteric alteration of the magnetic minerals, suggest that most of the tilting had occurred before the next normal polarity epoch. The magnetization of the approximately 26-m.y.-old Rio Hondo pluton, which indicates no significant tilting, also suggests that it became magnetized after the tectonic activity had waned.

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Structural deformation, however, had ceased altogether by the time the 23-m.y.-old Sulfur Gulch, Bear Canyon, and Red River(?) intrusions along the southern caldera margin and the 22-m.y.-old Lucero Peak pluton to the south were emplaced and magnetized. Solidification of the subcaldera plutonic rocks after eruption of the Amalia Tuff effectively thickened the crust in this region, restricting further tilting in the vicinity by a similar style of extension, which continued to the north of the caldera, tilting the sedimentary rocks and 15-m.y.-old basalt flows near Amalia (Figure 3). In relation to a typical model of the caldera cycle [*Smith and Bailey*, 1968], extension appears to have coincided with the onset of resurgence within the Questa caldera.

The transition from extensional tectonics characterized by tilted, presently low-angle, normal faults to the more recent tectonic regime characterized by untilted, high-angle, normal faults occurred within the caldera structure around 23–25 m.y. ago, and to the north of the Questa caldera between the 15-m.y. age of the tilted sedimentary rocks and basalts and the 7- to 11-m.y. age of the untilted rhyolite dome. Continued cooling of the crust during the 20- to 13-Ma relative lull in volcanism along the Rio Grande rift [*Chapin and Seager*, 1975] allowed the younger, more northerly trending, high-angle, normal faults to form and penetrate to deeper levels of the crust. Uplift, without significant tilting of the rocks in the Questa region, has continued through Servilleta time to the present.

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